

▲ 1 Problem Statement

Data type: Colour (RGB) medical images captured by an otoscope – a handheld device that illuminates and magnifies the ear canal and tympanic membrane.

What is predicted, from what: Given a single otoscopic image, the model predicts one of **8 clinical categories**: Normal, Acute Otitis, Serous Otitis, Retraction, Perforation, Serous Otitis Retraction, No Tympanic Membrane Visible (NoTM), and Ventilation Tube. A secondary binary task (Healthy vs. Diseased) is also formulated.

Why it matters: Otitis media and related conditions are among the most common causes of antibiotic prescription and pediatric hearing loss worldwide. Accurate diagnosis from otoscopic images requires years of clinical training; access to ENT specialists is severely limited in many regions. A reliable Computer-Aided Diagnosis (CAD) tool can assist primary-care clinicians, reduce misdiagnosis, and enable tele-medicine workflows in low-resource settings.

▲ 2 Dataset Registration

Source	Private clinical collection annotated by ear, nose and throat (ENT) specialists; accessed through EravciLab, TOBB University of Economics and Technology, Ankara, Turkey.
Licence	Institutional / private use – not publicly released.
Total samples	20,196 images (Train 15,598 • Validation 698 • Test 3,900)
Classes	8 – see distribution table below.
How obtained	Dataset provided by clinical collaborators of EravciLab with pre-defined patient-level train / validation / test splits to prevent data leakage between partitions.

Class	Train	Train %	Val	Val %	Test	Test %
Normal	2,002	12.8%	420	60.2%	2,335	59.9%
Acute Otitis	2,009	12.9%	43	6.2%	122	3.1%
Serous Otitis	1,895	12.1%	36	5.2%	177	4.5%
Retraction	1,917	12.3%	47	6.7%	186	4.8%
Perforation	1,987	12.7%	45	6.4%	402	10.3%
Serous Otitis Retraction	1,963	12.6%	40	5.7%	108	2.8%
NoTM	1,881	12.1%	46	6.6%	473	12.1%
Ventilation Tube	1,944	12.5%	21	3.0%	97	2.5%
Total	15,598	100%	698	100%	3,900	100%

Training set is near-perfectly balanced (~12% per class). Validation and test sets reflect the real clinical prevalence: Normal cases dominate (~60%), with several pathological classes at under 5%.

▲ 3 Feature Extraction Plan

Approach	Method	Rationale
Pretrained embedding (primary)	ResNet50 / ResNet18 features extracted from the penultimate layer after ImageNet fine-tuning. Input resized to 512×512.	Captures hierarchical colour, texture, and shape cues critical for discriminating subtle membrane changes.
Handcrafted features (baseline)	GLCM (texture energy, contrast, correlation), HOG (edge structure), LBP (local texture patterns); concatenated into a single feature vector.	Provides an interpretable baseline and shows the benefit of learned over engineered features.
Data augmentation	HorizontalFlip, ShiftScaleRotate ($\pm 15^\circ$), RandomBrightnessContrast, Gaussian/MotionBlur (via Albumentations). ImageNet normalisation.	Addresses limited intra-class variability and prevents overfitting on the balanced training split.

■ 4 Evaluation Metric

Primary metric: Weighted (Balanced) Accuracy — the macro-average of per-class recall, i.e. the mean of sensitivity across all 8 classes.

Why not plain accuracy? In the test set the Normal class holds ~60% of samples. A trivial classifier that predicts “Normal” for every image achieves 60% accuracy while completely failing to detect any disease. Weighted accuracy penalises this equally across all classes, regardless of their prevalence, and is therefore the only metric that faithfully measures diagnostic capability for minority pathological classes (e.g. Serous Otitis Retraction at 2.8%, Ventilation Tube at 2.5%).

Secondary metric: macro-F1 score — reported alongside weighted accuracy to capture the precision–recall trade-off per class. Together these two metrics give a complete picture of model performance under the severe real-world class imbalance present in the evaluation sets.

This proposal was prepared for the ML/DL course dataset registration (EravciLab, TOBB ETU, October 2026). The dataset is a private clinical collection and has not been publicly released.